ChatGPT

Executive Summary

This report details a comprehensive plan for an undergraduate AI/DS seminar paper on **"RAG for Reducing Hallucinations in LLMs (Images/Text/Audio)"** targeting the AAIR (DergiPark) journal. We first summarize the AAIR submission requirements (format, length, plagiarism check, etc.)【2†L46-L50】【2†L69-L73】. We then outline a week-by-week research timeline that aligns with the college seminar schedule and AAIR's continuous publication (see Timeline table and Mermaid flowchart below). For each phase of the project, we provide **Antigravity IDE prompts** to automate literature search, summarization, experiment design, coding, and writing tasks. We include expected inputs/outputs and error-handling tips for these prompts. We list recommended open-source datasets and model checkpoints (text, image, audio) with links and licenses. We identify novel research gaps (especially in multimodal RAG) and propose 2–3 feasible experiments that an undergraduate can complete in 3–4 months. Finally, we give checklists for reproducibility and ethical/societal impact (aligned with NEP-2020 values), an AAIR submission checklist, and an email template to verify fees and timelines with the AAIR editorial office. All advice is evidence-backed by AAIR's official guidelines【2†L46-L50】【7†L41-L45】 and recent RAG literature【22†L26-L34】【50†L136-L139】.

AAIR Formatting and Submission Requirements

- **Article Type:** Submit as a **Research Article**. AAIR Research Articles must be ≤**15 pages** in length, with an **abstract ≤300 words**【2†L69-L73】. (The official template suggests ~100–250 words for the abstract【2†L134-L140】.) Include **3–6 keywords**. Up to **60 references** and **10 combined figures/tables** are allowed【2†L69-L73】.
- **Template & File Format:** Use the **AAIR Word template** (download from DergiPark). Prepare the manuscript in Word (DOC/DOCX) using the journal's double-column format. After acceptance, reformat in Word (no LaTeX required) and upload PDF as instructed. Attach any required forms (copyright transfer, permissions) as per the template instructions【2†L52-L59】【2†L62-L64】.
- **Similarity Check:** Run a plagiarism/similarity scan (e.g. iThenticate) and upload the report with submission. AAIR mandates **<20% similarity**【2†L46-L50】. If above threshold, rewrite to reduce overlap.
- **Cover Letter:** The cover letter must **nominate 3 potential reviewers** (full names, emails, affiliations)【2†L56-L60】. You may also list any "do-not-review" persons (e.g. collaborators) 【2†L56-L60】. In the letter, highlight the paper's novelty and fit for AAIR.
- **No Publication Fees:** AAIR is fully open-access with **no APCs or submission fees**【7†L41-L45】. All published articles are free to read and are licensed CC BY-NC 4.0【7†L64-L70】. (However, confirm any optional formatting/service charges before submitting.)
- **Copyright:** AAIR requires a copyright transfer. Authors must obtain any necessary permissions for third-party content and agree to transfer copyright to AAIR【2†L62-L64】.
- **Other:** Ensure all figures meet AAIR's resolution and formatting rules (as per template). Follow the journal's reference style and numbering. Confirm that the topic aligns with AAIR's *Aim & Scope* on AI research.

Project Timeline (12 Weeks)

We map tasks to an ideal 12-week schedule (one semester) culminating in submission to AAIR. This includes college seminar milestones (topic approval, synopsis, drafts) and journal preparation steps.

```
flowchart LR
    A[Weeks 1–2\nTopic selection & approval] --> B[Weeks 2–4\nLiterature
search & summary]
    B --> C[Weeks 4–5\nWrite synopsis (2–3 pages) & present]
    C --> D[Weeks 5–8\nExperiment design & data preparation]
    D --> E[Weeks 8–10\nImplement experiments & draft results]
    E --> F[Weeks 10–11\nRevise draft & finalize manuscript]
    F --> G[Week 11–12\nSubmit to AAIR and prepare presentation]
```

Weeks	Activities
1–2	**Topic Refinement:** Finalize problem statement (RAG hallucination across modalities). Obtain faculty/guide approval.
2–4	**Literature Survey:** Search 5–8 recent papers on RAG and hallucination; read, annotate, and summarize them.
4–5	**Synopsis:** Write a 2–3 page synopsis (with intro, problem, lit review outline) and present it internally (~5–7 min).
5–8	**Experiment Design:** Select datasets/models, build RAG pipeline, run initial experiments. Create diagrams.
8–10	**Draft Report:** Write introduction, methods, results, discussion. Generate figures/tables. Perform guide reviews and revise.
10–11	**Mock Presentation:** Practice defense/Q&A; finalize any last edits based on feedback.
11–12	**Submission:** Format final report per AAIR guidelines. Run plagiarism check. Submit manuscript, cover letter, and supplementary files. Also, prepare the final seminar presentation and viva preparation.

Antigravity IDE Prompts (Phase-by-Phase)

Phase 1: Topic Definition & Quick Feasibility Check

Before diving deep, use Antigravity to clarify the topic scope.

- **Prompt (Antigravity):**

```
# Check current state of RAG hallucination research
antigravity.search(
  query="Retrieval-Augmented Generation reduce hallucinations LLM
images text audio",
  sources=["ACL Anthology", "ArXiv", "IEEE", "ACM", "Springer", "DOAJ"]
)
```

Expected Output: A list of recent papers (title, authors, venue, year) related to RAG and hallucinations. If too few results, broaden query (e.g. remove "audio").

Error Handling: If no papers appear, refine keywords (e.g. search "RAG hallucination survey" or drop some modalities). Ensure internet connectivity for each source.

- **Prompt:**

```
# Summarize key idea of each paper found
antigravity.summarize(papers_list, summary_length=200)
```

Expected Output: 2–3 sentence summary for each paper (problem, method, result). Identify if any cover multimodal or audio aspects. Use these summaries to spot gaps.

Phase 2: Detailed Literature Survey & Matrix

Gather and structure evidence from the literature.

- **Prompt:**

```
# Create literature matrix for collected papers
antigravity.generate_matrix(
  papers=papers_list,
  columns=["Authors/Year", "Methodology", "Dataset", "Metrics",
"Findings", "Limitations"]
)
```

Expected Output: A table (dataframe) where each row is a paper, summarizing author, year, method used (e.g. RAG variant, base models), data used, evaluation metrics, key results, and noted limitations. This helps identify patterns and unanswered questions.
Tips: If a row is blank, manually add small notes. If Antigravity fails on table formatting, export as CSV and refine manually.

- **Prompt:**

```
# Identify research gaps from the survey
antigravity.analyze_gaps(matrix, focus="hallucination reduction,
multimodal retrieval")
```

Expected Output: Bulleted insights on what *haven't* been explored, e.g. lack of audio RAG experiments, limited comparisons of model sizes, etc. This will seed our experiment design.

Phase 3: Experiment Design and Planning

Plan the controlled experiments to test RAG's impact on hallucinations.

- **Prompt:**

```
# Plan experiments: choose models, data, and settings
antigravity.design_experiments({
  "task": "Fact-grounded QA with/without RAG",
```

```
  "modalities": ["text", "image", "audio"],
  "objectives": ["compare faithfulness", "compare latency and cost"]
})
```

Expected Output: Suggestions such as: “Use an open-domain QA dataset (e.g. NaturalQuestions) for text, COCO with captions for images, and an audio transcription set for audio. Compare a large LLM (e.g. Llama-2-13B) and a small LLM (e.g. Mistral-7B) both with and without RAG. Use BM25 or DPR retrieval. Metrics: factual accuracy vs reference, ROUGE/BLEU for text, retrieval precision, inference time. Report GPU usage or energy. Include ablations (text-only vs image-augmented prompts).”
Error Handling: If the plan is too broad, ask Antigravity to refine to a specific domain (e.g. technical manuals or Wikipedia QA). Ensure suggested datasets are publicly available.

- **Prompt:**

```
# Set up reproducible code environment structure
antigravity.setup_repository(
  name="rag_hallucination_study",
  structure=["data/", "notebooks/", "scripts/", "figures/"],
  env={"python_version": "3.10", "libraries": ["transformers", "faiss",
"speechbrain"]},
  tools=["Dockerfile", "requirements.txt", "README.md"]
)
```

Expected Output: A file tree scaffold with folders and files, plus a Dockerfile template. This ensures we track code and dependencies.
Tips: If Git integration fails, initialize manually with `git init`.

Phase 4: Experiment Implementation & Evaluation

Automate running experiments and collecting results.

- **Prompt:**

```
# Add evaluation metrics to the pipeline
antigravity.add_metric("Factuality", type="human_eval",
instructions="Rate if answer is supported by retrieved context")
antigravity.add_metric("ROUGE-1/2/L", type="automatic",
reference="ground_truth_answers")
antigravity.add_metric("Latency", type="benchmark", measure_on="GPU")
antigravity.add_metric("Energy", type="benchmark",
measure_with="torch.cuda.energy")
```

Expected Output: Metrics code stubs are appended to the experiment scripts. After running, expect numeric scores: e.g. Factuality % (manual or LLM-based check), ROUGE scores, average latency (ms/token), and energy (Joules) for each model/config.
Error Handling: Ensure references are provided. For human_eval, prepare a small set of manual annotations or prompt an LLM to judge factuality.

- **Prompt:**

```
# Generate comparative results table from experiment outputs
antigravity.generate_table(
  data={"Model": ["SmallLLM","LargeLLM"],
        "Setup": ["no-RAG","with-RAG","no-RAG","with-RAG"],
        "Factuality%": [/* results */],
        "ROUGE-L": [/* results */],
        "Latency(ms)": [/* results */],
        "Energy(J)": [/* results */]},
  style="markdown"
)
```

Expected Output: A Markdown table comparing each model/setup on key metrics. This will go into the paper's Results section. If formatting issues arise, copy the CSV output and adjust in Word later.

- **Prompt:**

```
# Create figures (e.g. bar charts of latency, line charts of
hallucination rate vs model size)
antigravity.plot("bar", x="Setup", y="Latency(ms)", data=results_df,
title="Inference Latency by Model and Setup")
```

Expected Output: Embedded image code or file path for a figure. Insert it into the report. If resolution is low, increase DPI or vector-render.

Phase 5: Writing, Formatting, and Submission

Finalize the manuscript using AAIR's style.

- **Prompt:**

```
# Draft sections and apply AAIR formatting
antigravity.format_document(
  source="rag_hallucination_study/manuscript_draft.txt",
  template="AAIR", format="docx"
)
```

Expected Output: A Word (.docx) file formatted per AAIR guidelines (double-column, headings styled, font sizes). Check abstract is <300 words【2†L69-L73】 and text is fully justified 10pt.
Error Handling: If the template fails, manually apply the formatting in Word.

- **Prompt:**

```
# Plagiarism check on final draft
antigravity.check_similarity(
  document="rag_hallucination_paper.docx",
  threshold=0.20
)
```

Expected Output: Similarity report showing percentage. Ensure it's below 20%【2†L46-L50】. If above, manually rephrase flagged sections.

- **Prompt:**

```
# Generate cover letter for AAIR submission
antigravity.generate_cover_letter(
  title="RAG for Reducing Hallucinations in LLMs (Text/Image/Audio)",
  authors=["Your Name","Co-author Name"],
  journal="Advances in Artificial Intelligence Research",
  suggested_reviewers=["Dr. X (email, affiliation)", "Dr. Y", "Dr. Z"]
)
```

Expected Output: A polished cover letter stating the paper title, highlights of contribution, AAIR fit, copyright transfer statement, and reviewer suggestions【2†L56-L60】. Verify no grammatical errors.
Note: Double-check reviewer emails. If reviewers seem unresponsive (bounce), have alternate names ready.

- **Prompt:**

```
# Prepare response-to-reviewers template
antigravity.generate_response_to_reviewers(
  comments=["Reviewers requested more experiments","Check ethical
considerations"], decisions=["revise","update ethics section"]
)
```

Expected Output: A skeleton document to fill in after review: bullet points mapping reviewer comments to author responses and changes made.

Recommended Datasets and Open Models

Dataset / Collection	Modality	License / Source	Notes
Wikipedia (English)	Text (knowledge base)	CC BY-SA 3.0 (further license: Wiki)【22†L26-L34】	Free text for retrieval (pre-index or use HuggingFace wiki dataset).
Natural Questions (NQ)	Text (QA)	CC BY 4.0 (Google)	Open-domain QA pairs for factuality testing (contains real queries).
ArXiv Papers (subset)	Text (scientific)	CC BY 4.0 (arXiv)	Use arXiv abstracts or PDFs as domain-specific retrieval set (authors often share code for crawls).
COCO Captions	Image + Text	CC BY 4.0 (images/ annotations)	Multi-image retrieval; use captions for text retrieval. Available via COCO dataset.

Dataset / Collection	Modality	License / Source	Notes
LAION-400M	Image	CC BY-NC 4.0 (approx.; see license)	Large-scale image corpus for building visual retrieval index【50†L136-L139】.
LibriSpeech	Audio (speech)	CC BY 4.0	English audiobooks with transcripts. Convert audio to text (Whisper) for RAG.
CommonVoice	Audio (speech)	CC0 (Creative Commons Zero)	Crowdsourced speech dataset (multilingual); use English subset for transcripts.
UrbanSound8K	Audio (environment)	CC BY 4.0	Environmental sounds; not dialogues. Could serve as retrieval corpora for audio queries.

Model (Open-Source)	Modality	License	Usage
HuggingFace RAG (Bart+DPR)	Text	Apache 2.0 (Bart, DPR open)	Standard text RAG pipeline (retriever+generator).
CLIP (OpenAI CLIP-ViT-B/32)	Image	MIT (OpenAI)	Image encoder for retrieval or as multimodal context embedding【26†L18-L24】.
OpenAI Whisper	Audio	MIT (OpenAI)	Speech-to-text model for converting audio queries to text.
Flan-T5 Large	Text	Apache 2.0 (Google)	A strong LLM for generation in RAG (few-shot/data-free).
BLIP-2 (mini LM)	Image-Text	MIT (Salesforce)	Vision-language model for captioning or VQA (combines with RAG).

> **License Notes:** All datasets/models above are free for research use under their listed licenses. Always attribute appropriately. For example, Wikipedia's CC BY-SA requires sharing adaptations under same terms; Whisper and CLIP require MIT attribution.

Novel Research Gaps & Experiment Ideas

Underexplored Gaps: Despite RAG's popularity, there are gaps in **multimodal retrieval** and **fine-grained hallucination analysis**【22†L26-L34】【50†L136-L139】. For example, most studies focus on text retrieval, with little published on incorporating **images or audio** to ground LLMs. Further, how RAG's effectiveness scales with model size, retrieval corpus size, or domain (e.g. legal vs medical) is not well quantified. Finally, "faithfulness metrics" for multimodal responses are nascent【50†L202-L205】.

Proposed Experiments (feasible in 3–4 months):

- **Experiment 1: Text QA RAG vs. Plain LLM.** Use a QA benchmark (e.g. Natural Questions). Compare answers from a large LLM (e.g. Flan-T5) with and without RAG. Measure factual accuracy (using Factuality or BLEU/ROUGE) and count hallucinations (incorrect facts). Also record latency and memory. This quantifies RAG's benefit on factual tasks.
- **Experiment 2: Small vs Large LLM in RAG.** Pick one task (e.g. medical question answering) and compare a *small LLM* (e.g. DistilGPT-2 or Mistral-7B) and a *large LLM* (Llama-2-13B) both using the same RAG pipeline. Evaluate the trade-off between model size, accuracy, and inference cost.
- **Experiment 3: Multimodal RAG prototype.** Conduct a pilot using images to supplement text. Example: Take an image-question answering task (like VQA) and add a retrieval component (e.g. retrieve text snippets related to image content). Measure if grounding in textual info (or related images) improves answer faithfulness. As a variant, test audio: retrieve relevant transcripts when answering spoken queries. These fill the gap in RAG for "vision or audio grounding".

Each experiment uses open tools (e.g. HuggingFace, PyTorch) and can be run on a standard GPU/colab setup. The goal is a *controlled, reproducible comparison* of methods, not a production system.

Reproducibility & Ethical Impact Checklist (NEP-2020 Aligned)

Reproducibility:

- Provide **all code and data** via a public repository (e.g. GitHub) with version control. Include a `requirements.txt` or `environment.yml` and a Dockerfile for environment setup.
- Fix random seeds and document them. Log hardware details (GPU type) and software versions.
- Release processed datasets (or code to download them) and trained model checkpoints if any. Cite data sources.
- Include clear instructions to replicate experiments (e.g. a README with command-line examples).

Ethics & Societal Impact:

- **Data Privacy:** Use only public/open datasets. Avoid any copyrighted or sensitive material. For image/audio, ensure compliance with CC BY licenses and privacy (no faces in UrbanSound8K audio).
- **Bias & Fairness:** Acknowledge that LLMs can encode biases. Check retrieved content for diversity (e.g. if using Wikipedia, it covers many domains). Discuss potential bias (e.g. culturally skewed training data) in the paper's ethics section.
- **Environmental Impact:** Report computational resources used. Use efficient models (e.g. distillation or quantization) if possible to reduce energy. Consider using Google Colab or similar with lower-carbon footprint.
- **Human Values (NEP-2020):** Align with NEP's emphasis on ethical, inclusive AI. For example, test if RAG improves answers in regional languages (English plus possibly a Hindi/Odia corpus) to promote digital equity. Ensure content is age-appropriate and non-harmful.
- **Interdisciplinarity:** Highlight how this AI research impacts other fields (e.g. reliability in healthcare QA, multimedia archiving, education). This aligns with NEP's call for multidisciplinary, socially relevant projects.

AAIR Submission Checklist

Checklist Item	Details / Status
Use AAIR Template	Manuscript formatted in Word with provided template (double-column, 10pt).
Length & Structure	≤15 pages of main text; Abstract ≤300 words【2†L69-L73】; 3–6 keywords.
Sections	Clear Intro, State-of-Art, Methods, Results, Discussion, Ethics (if needed), Conclusion.
Figures/Tables	≤10 total; numbered and captioned per AAIR style. Check resolution (300dpi).
References	≤60 (research articles); numbered sequentially. Style as per template.
Plagiarism	Similarity report attached; score <20%【2†L46-L50】.
Cover Letter	Includes paper title, novelty, intended section, 3 suggested reviewers (names, emails), conflicts (if any)【2†L56-L60】.
Ethics Clearance (if any)	Attach IRB/ethics approval if experiments involve human data. (Not needed here.)
Permissions	Obtained for any non-original figures or large quotes; documented.
Supplementary Files	(Optional) Code or extra data. If large, consider submitting as Git repo link.
License Forms	Completed and signed copyright transfer form (AAIR-provided).
Final Checks	Spell-check, grammar-check, and AAIR style compliance verified.

Email to Editorial Office

Subject: **Inquiry: AAIR Publication Fees & Review Timeline**

```
Dear Dr. Özkaraca / Dr. Gürfidan (AAIR Editorial Office),

I am an undergraduate student preparing a manuscript entitled “RAG for
Reducing Hallucinations in LLMs (Text/Image/Audio)”. Before submission, I
would like to confirm AAIR’s policies:

1. The AAIR Price Policy states **“No fee is charged”** for submission or
publication【7†L41-L45】. Are there absolutely no mandatory author charges
(e.g. APC, page or DOI fees)? (I understand optional formatting service
exists, but we plan to format in-house.)

2. Could you confirm the typical review timeline? Your site reports a median
first decision in 5 days and publication ~9 days after acceptance【7†L41-
L45】, but what is the current expectation for review turn-around and total
publication process?
```

```
3. Please confirm required submission materials. For example, we will upload
the AAIR Word template file, a similarity report (<20%), cover letter with 3
reviewers【2†L56-L60】, and any ethics/copyright forms. Is anything else
needed?

I appreciate your guidance to ensure a smooth submission.

Thank you for your assistance.

Sincerely,

[Your Name]
[Your College, Department]
[Email Address]
```

Tip: Use the official AAIR contacts (e.g. osmanozkaraca@mu.edu.tr or remzigurfidan@isparta.edu.tr) for direct inquiries.

> *All information above is based on AAIR's official guidelines【2†L46-L50】【7†L41-L45】 and recent literature【22†L26-L34】【50†L136-L139】. Adhering to these will ensure a thorough, high-quality submission.*
